

LINK TO CODE: https://colab.research.google.com/drive/1rW12Wyo_wLB2PWk-fnLghUtoNnLDVvFb#scrollTo=6tOgav3tF8jT
LINK TO DATA SET: <https://www.kaggle.com/datasets/humairmunir/anaemia-prediction-dataset>
LINK TO FOLDER: <https://drive.google.com/drive/folders/1m7TKIfDSAsmml-xvUMTwPZQqv-q09nG->

Predicting Anemia Using RGB Pixel Values from Images of Patients' Blood and Machine Learning

Hiranya Sundar, Nathan Gong, Ali Moazzami, Princeton Liu, Abdallah Fares, UCLA

ABSTRACT

Anemia is a widespread blood condition that disproportionately affects members of the Global South. The difficulty and cost of diagnosis, especially for under-resourced regions, limits the path to treatment for many of those affected with the condition. Creating a machine learning algorithm to rapidly, inexpensively, and accurately predict the presence of anemia can be vital for the wellbeing of those in underserved communities. We trained several machine learning algorithms, including logistic regression models, SVMs, and neural networks, to be able to predict a patient's anemia diagnosis based solely on the patient's sex and the pixel values obtained from an image of the patient's blood. Hyperparameter tuning was used to further increase model accuracy. A neural network model with 7 nodes and reLU activation function seemed most effective, with a 94% accuracy. The creation of a deep learning model with the capability to accurately and efficiently diagnose the presence of anemia decreases the resources needed for diagnosis significantly.

1 MOTIVATION

Anemia is a condition where either the amount of red blood cells or the amount of hemoglobin within those red blood cells falls below healthy levels [2]. This condition is largely a result of inadequate nutrient consumption (particularly iron consumption), infections including malaria and HIV, and genetic inheritance. In 2021 almost 24.3% of the world population [6], nearly 2 billion people, was infected with anemia, making it one of the most common ailments on the planet. Anemia disproportionately affects members of the Global South, primarily South Asia and Sub-Saharan Africa [2], jointly due to the prevalence of the aforementioned risk factors and poor medical infrastructure and access. Traditional anemia diagnostics, such as complete blood count, reticulocyte count, sonograms, and other tests and screenings [3], require complex and expensive lab equipment as well as the supervision of medical professionals, often making access to more remote and poorer populations, who are also the most at risk, prohibitively expensive.

To help solve these issues of cost and access we propose several machine learning algorithms trained on a dataset of images of anemic and non-anemic patients' blood that can correctly predict with a high accuracy the likelihood that a patient has anemic. Requiring only a properly-formatted image of a patient's blood, these algorithms will help significantly cut down the time, labor, and cost of diagnosing anemia in a standard lab and, for more remote and at-risk regions, widely increase accessibility to diagnosis, making later treatment more feasible. We train several different algorithms (logistic regression, SVM, and neural networks) implemented in Python and compare their efficacy against one another to determine the most accurate one that, as is important in any medical diagnostic method, also produces the least number of false negatives.

2 PROBLEM DEFINITION

We seek to predict whether a patient is anemic or not in a binary classification task, using patient gender and the RGB values taken from a picture of a patient's blood sample. By making anemia diagnosis more accessible and affordable, this project has the potential to save lives, particularly in underserved communities where traditional diagnostic resources are scarce.

3 PRE-PROCESSING & VISUALIZATION

Data from Kaggle [1] included numerical variables such as percentage of red, green, and blue pixels, and categorical variables like sex and anemia status (yes or no anemia presence). To prepare this data for analysis, categorical variables were converted into numerical format using one-hot encoding, missing values were

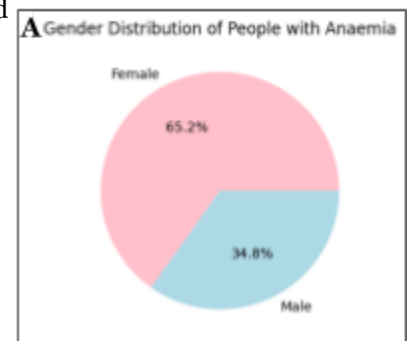


Fig. 1. Gender distribution of people with anemia

imputed with the KNN imputation method using five nearest neighbors, and all numerical values were standardized to a scale between 0 and 1.

A preliminary analysis of the dataset revealed significant gender disparities in anemia prevalence, with women exhibiting a higher incidence compared to men (Fig. 1A). Furthermore, distinct patterns emerged in the distribution of red, green, and blue pixel percentages across patient groups. As expected, red pixel values consistently exceeded those of green and blue due to the blood's inherent color composition. However, a notable observation was the higher red pixel percentage in non-anemic individuals, which aligns with the physiological understanding of reduced red blood cell count in anemia that contribute to the red hue of human blood.

Subsequent exploratory data analysis uncovered potential relationships between gender, RGB values, and anemia status. The overall RGB distribution comparing pixel percentages of red green and blue pixels found in the blood sample confirms the dominance of red and its association with anemia status as well as lower amounts of red in anemic patients (Fig. 2A). Specifically, there appears to be a clear reduction in red pixel percentage among anemic patients (Fig. 2B). There is also higher variability in blue pixel values for anemic individuals, as indicated by the increased interquartile range and presence of outliers (Fig. 2C). There also appears a subtle elevation in green pixel percentage within the anemic group (Fig. 2D).

These findings provide a foundation for further investigation into the potential of RGB color analysis as a diagnostic tool for anemia. Visible disparities between pixel color percentages and sex versus anemia presence provides the motive that a machine model might be able to predict anemia.

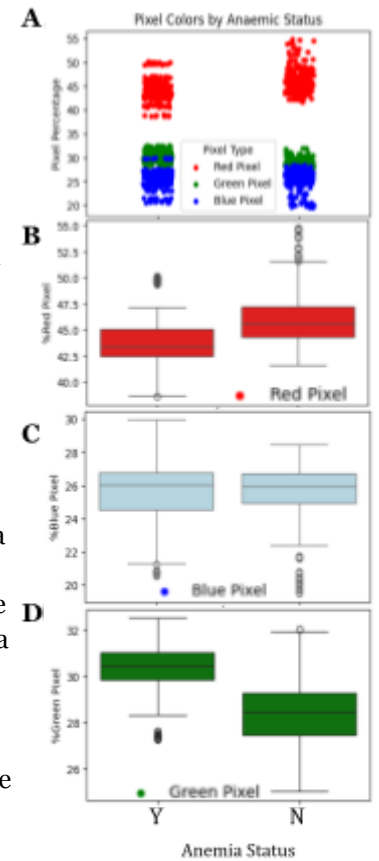


Fig. 2. Visualization of different metrics in relation to anemia diagnosis: **A.** Pixel Colors by Anemic Status **B.** % Red Pixel by Anemic Status **C.** % Blue Pixel by Anemic Status **D.** % Green Pixel by Anemic Status

4 MODELS

I. Logistic Regression

i. METHODS

The first algorithm tested was a standard logistic regression model in the scikit-learn Python package. Important features were visualized using the seaborn's sns feature importance functions, sklearn's logistic regression library was employed to establish a relationship between the input features and the target variable, which indicates the presence or absence of anemia. Hyperparameter tuning was conducted to optimize model performance. The regularization parameter, C, was varied across multiple values ranging from 0.01 to 100 to identify the optimal setting. The model was evaluated using a range of metrics including accuracy, precision, recall, F1-score, ROC AUC, false positive rate, and false negative rate.

ii. EXPERIMENT DESIGN & EVALUATION

We found that the most important features in the model were primarily the Sex of the patient and their image's %Green pixel value, though all features held significant weight (Fig. 3A). The weight of the %Green pixel value in particular suggests that there is something inherent in anemic vs. non-anemic blood that causes a shift in the amount of green imaged, a relationship which could be investigated further. The coefficient for sex is slightly positive, indicating that being female could be associated with a slightly higher risk of anemia compared to males. However, the impact of sex appears to be less pronounced than the percentage of green pixels.

Even with hyperparameter tuning, the model performs relatively poorly, with a maximum 0.76 accuracy rate and a false negative rate of about 0.08. The area under the curve (AUC) of the

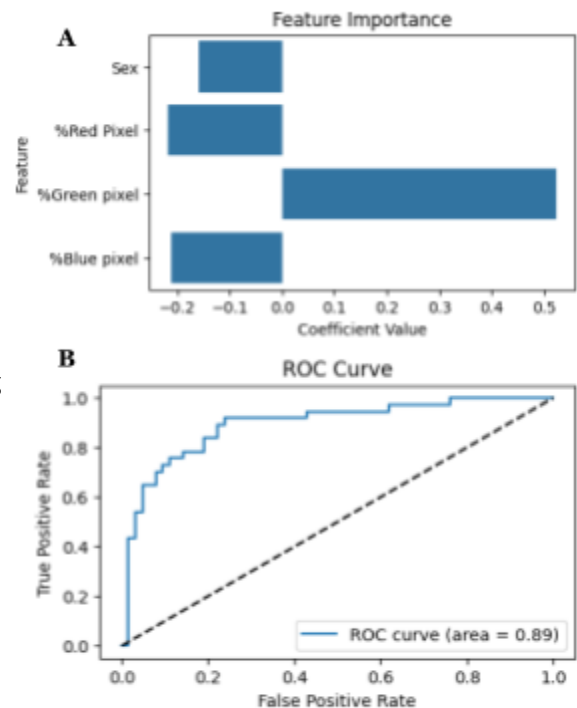
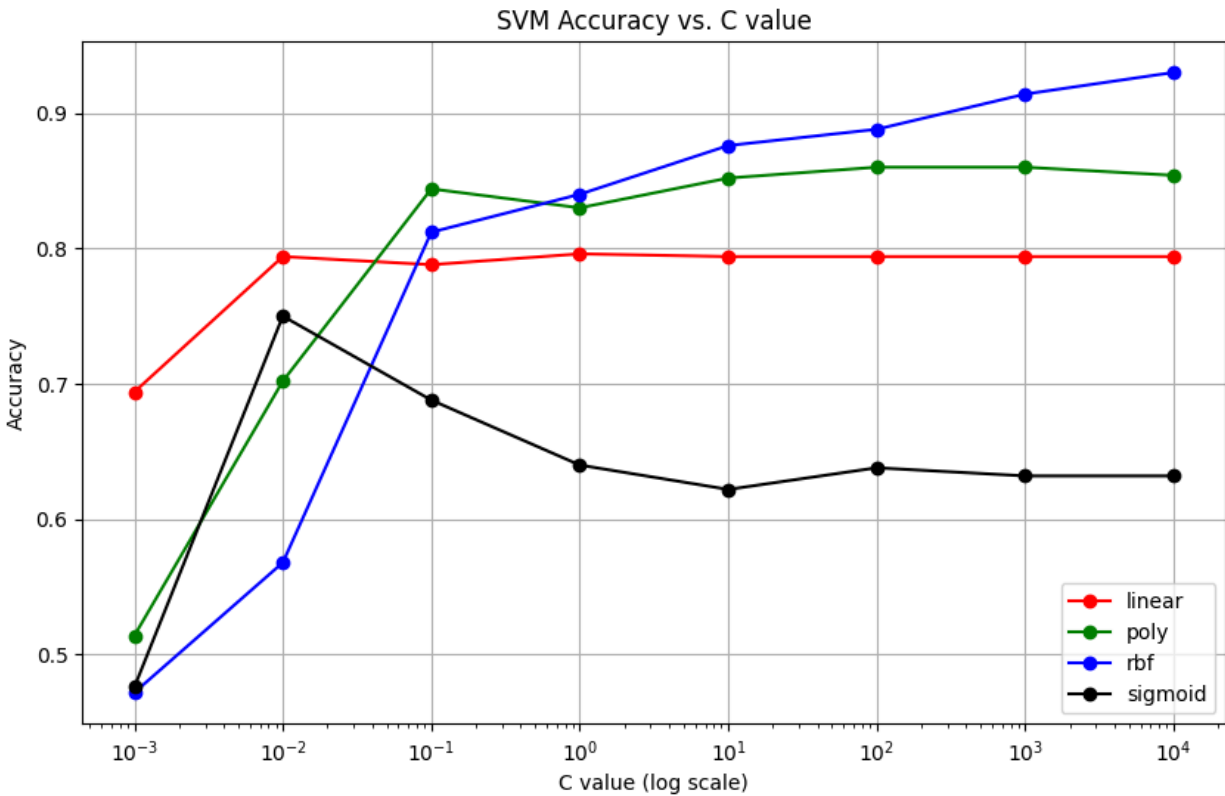


Figure 3. **A.** Graph of relative feature importance **B.** ROC curve for logistic regression model

ROC curve is 0.89, somewhat promising though still not enough for truly accurate diagnosis.

II. Support Vector Model



a. Linear

i. METHODS

A linear Support Vector Machine (SVM) model was implemented to predict anemia. Four important features used to model linear SVM were sex, and the percentage of red, green, and blue pixels in blood images. Before hyperparameter tuning and cross validation, a simple linear SVM model was trained with the linear kernel and default slack variable, C(regularization parameter). However, to select the best model, hyperparameter tuning was used on the linear SVM, trying different combinations of slack variables (ranging from 0.1 to 100) since the kernel parameter was fixed to be linear. This was done through using GridSearchCV method with 10 fold splits defined in its parameter to better evaluate the model. Having small enough hyperparameters to tune, this cross validation was fast and resulted in giving the best model with the best slack variable for linear SVM.

ii. EXPERIMENT DESIGN & EVALUATION

The linear SVM model achieved an accuracy of 0.88, with a false positive rate of 0.1633 and a false negative rate of 0.0784 (Fig. 4). These results indicate the model's ability to correctly classify most instances but with a higher risk of misclassifying anemic patients (false negatives) compared to non-anemic ones (false positives). Lowering the regularization parameter (C) implies that lambda is increased for the regularization parameter, which means the magnitudes of the parameters of the model will decrease. This is to help decrease overfitting by making the model less complex and decreasing its variance. However, lowering the regularization parameter only decreases the accuracy for training and testing for SVM's. This is likely because the SVM isn't nearly as susceptible to

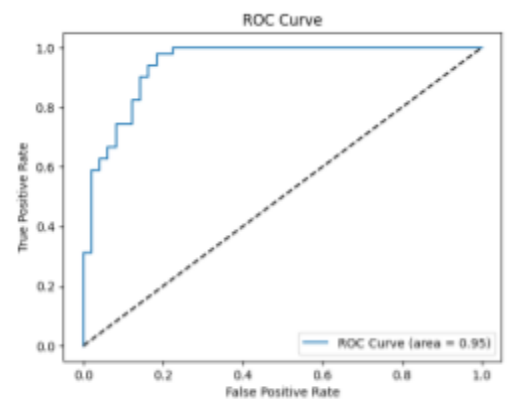
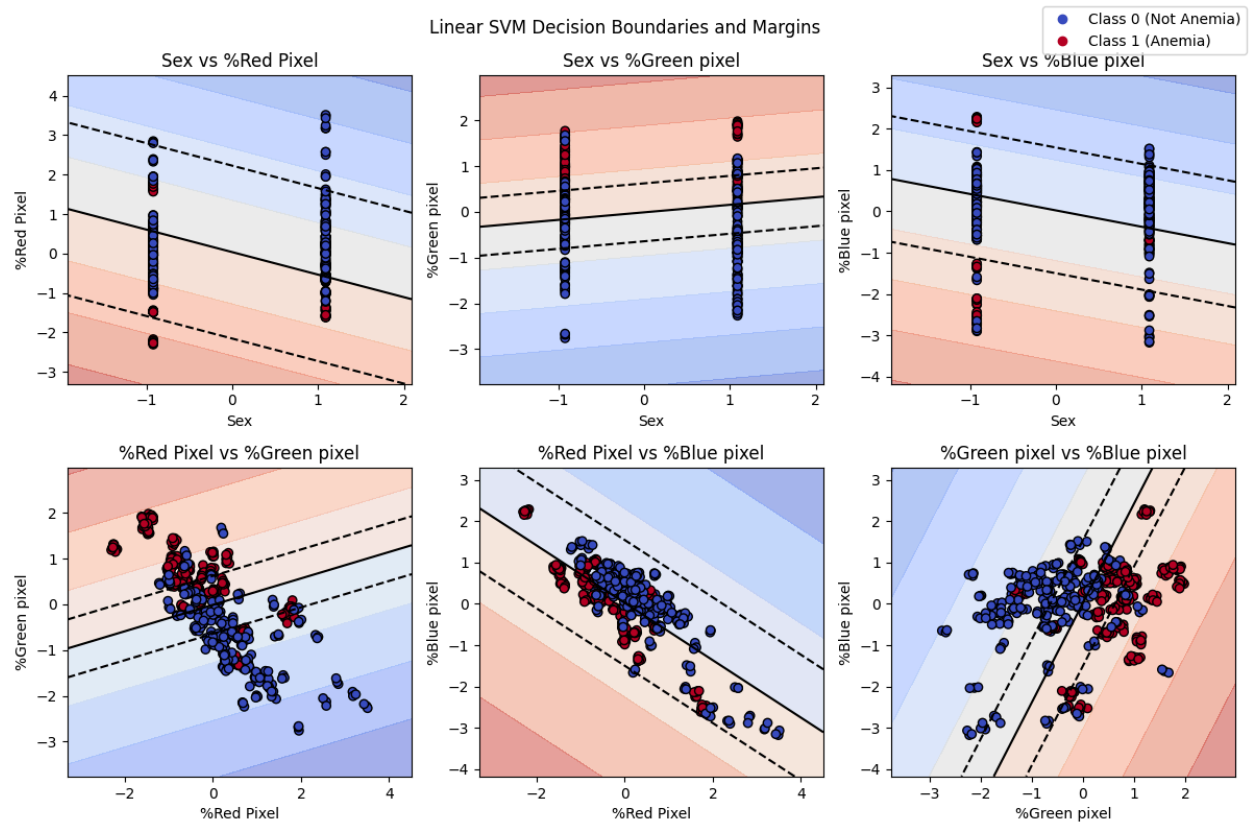


Figure 4. Linear SVM ROC curve

overfitting compared to complex models such as Neural Networks or polynomial regression because it maximizes the margins and doesn't have many parameters, which means each parameter is sensitive to change and vital to allowing the margin to perform accurately.



b. Non-linear

i. METHODS

A non-linear SVM model was explored to potentially capture more complex relationships in the data compared to the linear SVM. We used with 10-fold cross-validation was used to find the optimal combination of hyperparameters, including the kernel function (polynomial, rbf, sigmoid), the regularization parameter (C), and the gamma parameter (specific to non-linear kernels). The search aimed to identify the kernel type and hyperparameter settings that yielded the best performance.

ii. EXPERIMENT DESIGN & EVALUATION

The non-linear SVM achieved an accuracy of 0.94, surpassing the linear SVM. Additionally, the false positive rate (0.0816) and false negative rate (0.0392) were both lower, indicating superior performance in correctly classifying both anemic and non-anemic patients (Fig. 5). These results suggest that the non-linear SVM effectively captured the non-linear patterns within the blood image data, leading to more accurate anemia prediction. The final model used the RBF kernel with a C value of 100. It's important to note that non-linear models can be computationally expensive to train compared to linear models.

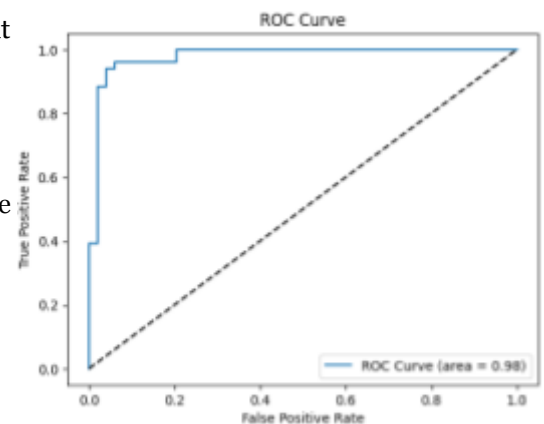


Figure 5. Non-Linear SVM ROC curve

III. Neural Networks

i. METHODS

A neural network model was developed to predict anemia based on four features: sex, and the percentage of red, green, and blue pixels in blood images. To optimize model performance, a hyperparameter tuning process was conducted, exploring various combinations of activation functions (ReLU, seLU, and tanh), layer sizes (1 - 10), and optimizers (adam, sgd, rmsprop). A total of 45 models were trained, selected randomly from a predefined set of hyperparameter values, exploring various combinations of activation functions, number of hidden layers, and optimizers. K-fold cross-validation was employed to assess model robustness. As this process was computationally intensive, with individual model training times exceeding an hour, particularly when combined with cross-validation, early stopping was implemented to decrease training time for future iterations to expedite the process.

ii. EXPERIMENT DESIGN & EVALUATION

The best model architecture was a 12 layer neural network, consisting of an input layer of 32 nodes. Additionally, 10 hidden layers had twenty 16 nodes each, and an output layer with two nodes for binary classification. The ReLU activation function was employed in the hidden layers to mitigate the vanishing gradient problem, while softmax was used in the output layer to generate class probabilities for the output layer. The optimal hyperparameters identified through this process included the Adam optimizer, which is well-suited for deep neural networks and can handle sparse gradients and noisy data. Cross-entropy loss was utilized as the loss function, and the model was trained for 50 epochs with a batch size of 10. K-fold cross-validation was employed to assess the model's robustness.

The resulting model demonstrated strong performance, achieving an accuracy of 0.935, precision of 0.886, recall of 0.994, F1-score of 0.937, and ROC AUC of 0.954 (Fig. 6). These metrics indicate high predictive power and reliability. However, the model's complexity and extensive training time raise concerns about potential overfitting and computational efficiency. We conclude the developed neural network model effectively predicts anemia based on the given features. While the model exhibits high performance, further optimization and evaluation are necessary to enhance its generalizability and efficiency. Incorporating early stopping and exploring alternative architectures could potentially improve training time and mitigate overfitting.

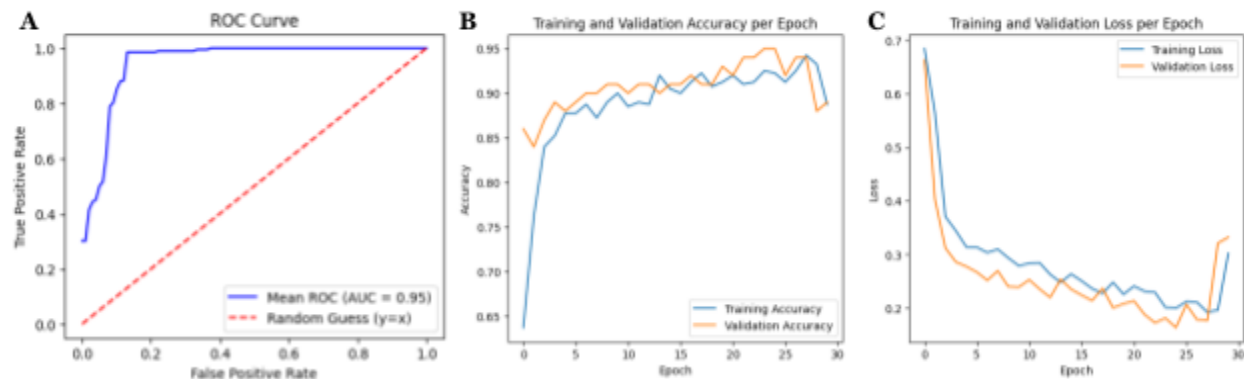


Figure 6. A. Neural network ROC B. Training and validation accuracy per epoch C. Training and validation loss per epoch

5 ANALYSIS AND RESULTS

The Neural Network model significantly outperformed the other models evaluated. It achieved an accuracy of [insert accuracy value], a false positive rate of [insert value], and a false negative rate of [insert value]. In comparison, Logistic Regression yielded the lowest accuracy of 0.77 and the highest false positive rate of 0.08. Linear and Non-Linear SVMs demonstrated intermediate performance, with accuracy values falling between those of the Neural Network and Logistic Regression. These findings clearly indicate that the Neural Network model is the most effective in accurately predicting a participant's anemia diagnosis based on the given dataset.

The superior performance of the Neural Network model can be attributed to its inherent capacity to capture both linear and complex nonlinear relationships within the data. In contrast, while Logistic Regression is suitable for linearly separable data, its performance was hindered by the underlying complexity of the anemia prediction problem. The linear SVM, despite its potential to handle non-linearity through kernel functions, was less effective than the Neural Network in this instance, suggesting that the data might not require the computational overhead of kernel-based methods. Further visualization of linear versus non-linear SVM decision boundaries proved invaluable, delineating how the relative 'linearity' of the data was unable to be handled by a non-linear SVM model.

Despite the increased accuracy and arguably superior performance of the Neural Network model, it has few drawbacks that need to be addressed. Deep learning neural networks are computationally intensive and susceptible to overfitting. Consequently, there is a trade-off between model complexity and performance, exacerbated by training and validation times when factoring in hyperparameter optimization and a stochastic k-fold validation model. Further research is warranted to explore strategies for enhancing the efficiency and generalizability of the Neural Network model while maintaining its predictive accuracy.

It is noteworthy that the inclusion of heme index, the diagnostic gold standard for anemia, initially resulted in perfect model performance. As low hemoglobin is how the presence of anemia is measured in the first place, it becomes evident why heme index and anemia presence provide a one-to-one relationship that skews the data. However, given the project's objective of evaluating the predictive capabilities of image-based features without relying on established diagnostic markers, the heme index was excluded from subsequent analyses. This decision allowed for a more realistic assessment of the model's potential in clinical settings where heme index might not be readily available.

6 DISCUSSION & CONCLUSION

The neural network model demonstrated superior performance in predicting anemia compared to logistic regression, linear SVM, and non-linear SVM. Its ability to capture complex patterns within the data was instrumental in achieving higher accuracy, precision, recall, and F1-score. While logistic regression and SVMs offer simpler and potentially faster alternatives, their performance was notably lower in this context. The computational demands of neural networks, particularly in terms of training time and hardware resources, present a significant challenge. While cloud-based platforms like Colab provide accessible computing power, offline environments may necessitate substantial investments in GPU hardware. This trade-off between computational efficiency and model performance is a critical consideration for practical applications.

Anemia is a global health concern, disproportionately affecting populations in developing countries with limited access to healthcare resources. Traditional diagnostic methods for anemia are often costly, time-consuming, and require specialized equipment and trained personnel. By leveraging image-based analysis and machine learning, this study explored the potential of developing an accessible and efficient anemia screening tool. The neural network model presented in this study represents a promising step towards this goal.

Future research should focus on refining the neural network architecture, exploring different feature engineering techniques, and evaluating the model's performance on larger and more diverse datasets. Additionally, investigating the integration of this model into point-of-care diagnostic devices could facilitate widespread implementation in resource-limited settings.

REFERENCES

1. "Anaemia Prediction Dataset." Kaggle: Your Machine Learning and Data Science Community, www.kaggle.com/datasets/humairmunir/anaemia-prediction-dataset.
2. "Anaemia." World Health Organization (WHO), 12 Nov. 2019, www.who.int/health-topics/anaemia#tab=tab_1.
3. "Anemia Screening - StatPearls - NCBI Bookshelf." National Center for Biotechnology Information, 25 July 2023, www.ncbi.nlm.nih.gov/books/NBK499905/.

4. Bajaj, Aryan. "Ensemble Models: How to Make Better Predictions by Combining Multiple Models with Python Codes...." Medium, 26 Apr. 2023, aryanbajaj13.medium.com/ensemble-models-how-to-make-better-predictions-by-combining-multiple-models-with-python-codes-6ac54403414e.
5. "A Free Online Introduction to Artificial Intelligence for Non-experts." Elements of AI, buildingai.elementsofai.com/Neural-Networks/from-logistic-regression-to-neural-networks.
6. The Lancet Hematology, GBD Anemia Collaborators. "Prevalence, years lived with disability, and trends in anaemia burden by severity and cause, 1990–2021: findings from the Global Burden of Disease Study 2021." The Lancet, [www.thelancet.com/journals/lanhae/article/PIIS2352-3026\(23\)00160-6/fulltext](https://www.thelancet.com/journals/lanhae/article/PIIS2352-3026(23)00160-6/fulltext).
7. "Medium." Medium, [meettank29067.medium.com/performance-measurement-in-logistic-regression-8c9109b25278https://meettank29067.medium.com/performance-measurement-in-logistic-regression-8c9109b25278](https://meettank29067.medium.com/performance-measurement-in-logistic-regression-8c9109b25278).

Task	People
Abstract	Abdallah and Hira
Motivation/Aims	Abdallah
Pre-Processing	Hira
Models - Logistic Regression	Abdallah (Hira contributed to code)
Models - SVM Linear/Non-Linear	Ali & Princeton
Models - Neural Networks	Hira (Princeton contributed to code)
References	Abdallah, Hira, Princeton, Ali, Nathan